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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

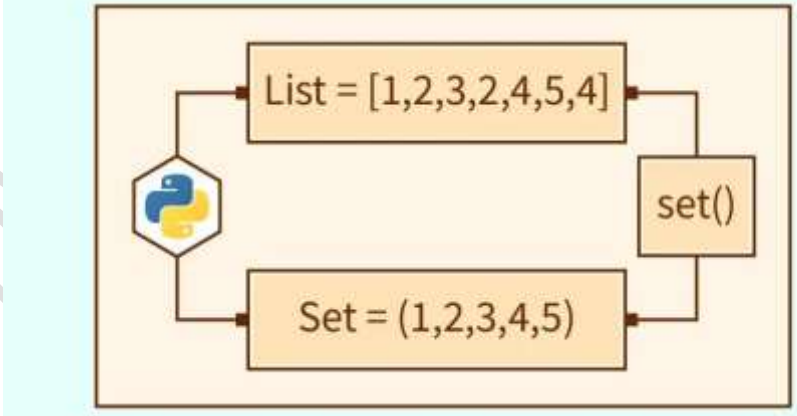
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 30: Sets in Python



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What is a Set?

A **Set** is a collection of **unique (non-duplicate)** items.

- Sets are written using **curly braces {}**.
- A set **does not allow duplicate values**.
- The order of items in a set is **not fixed**.
- Sets are useful for storing unique data.

Example

```
fruits = {"Apple", "Banana", "Mango"}

print(fruits)
```

Output

```
{'Apple', 'Banana', 'Mango'}
```

Note: The order of the items may appear differently each time you run the program.

Real-Life Example

Imagine a **class attendance list**.

If a student's name is entered twice:

- Rahul
- Anita
- Rahul
- Suresh

A **Set** automatically keeps only unique names:

- Rahul
- Anita
- Suresh

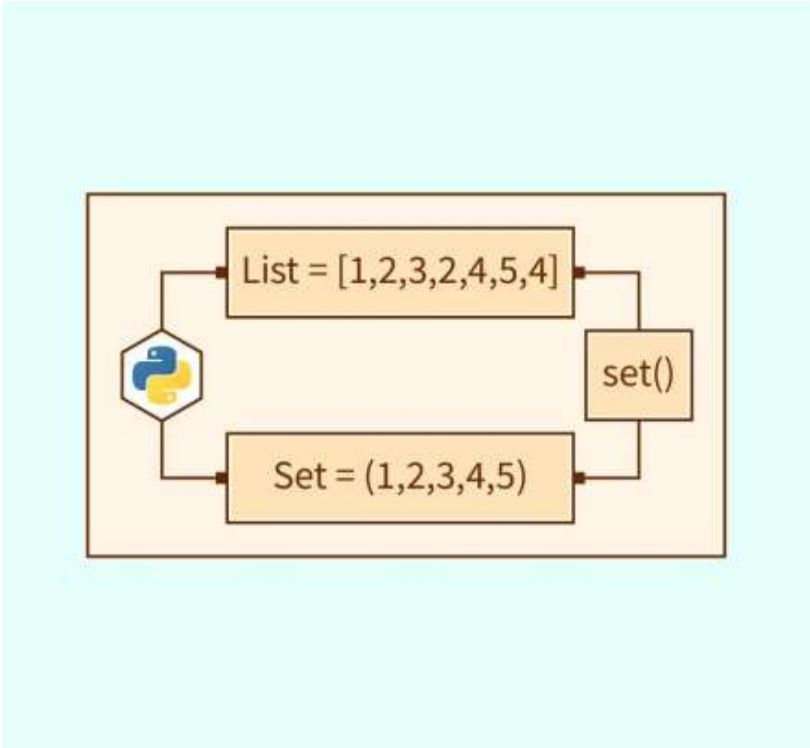
Features of Sets

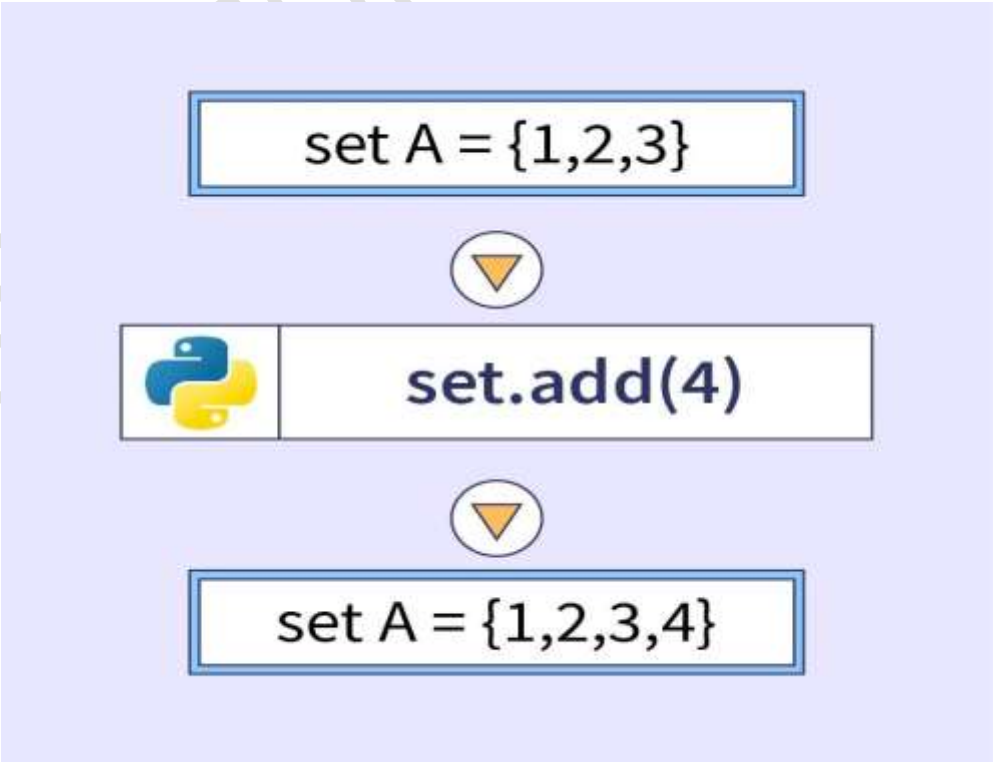
- Stores only **unique items**
- Removes duplicate values automatically

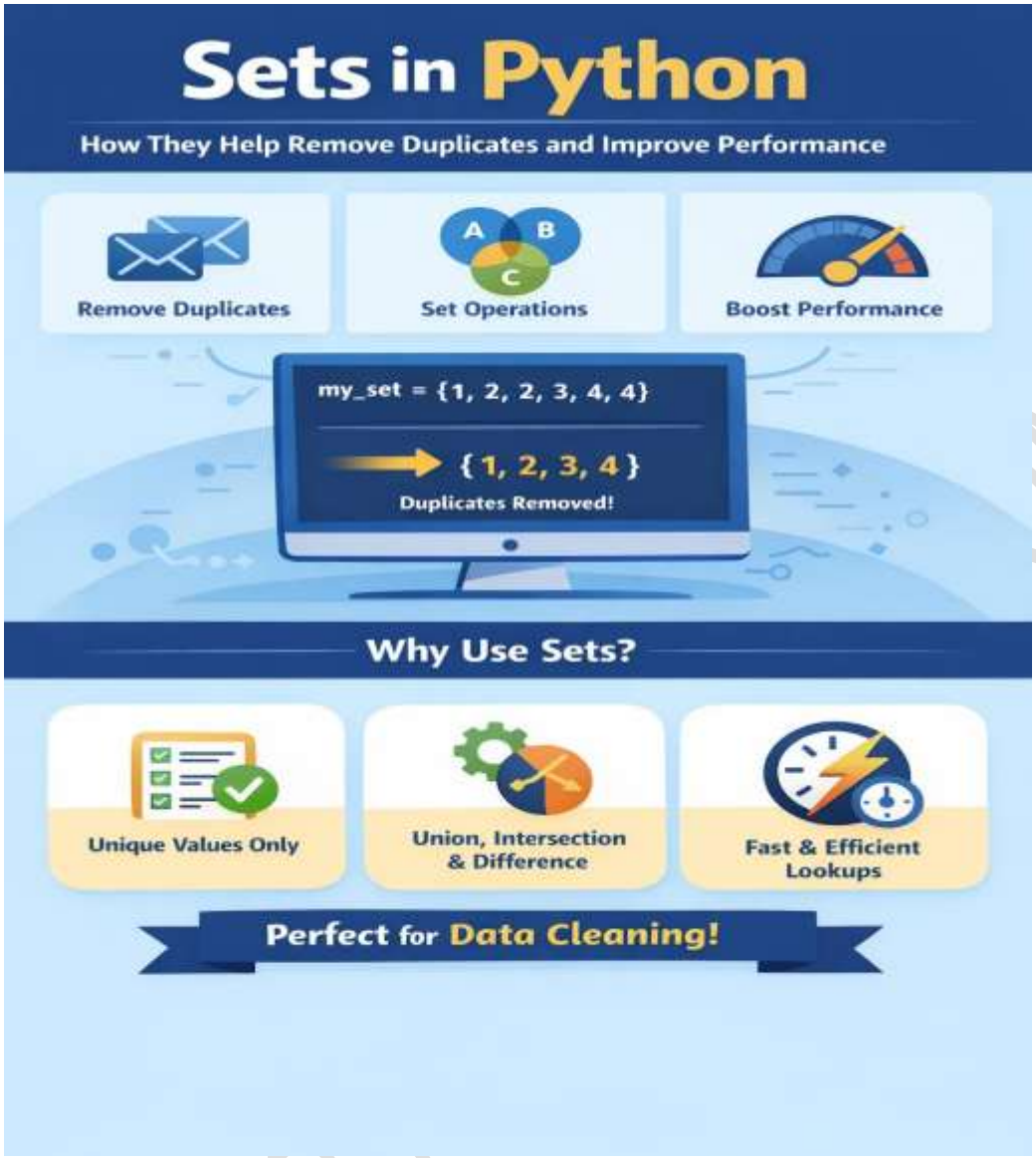
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- Unordered collection
- Mutable (you can add or remove items)
- Fast searching and checking

Set Structure







Set

{ Apple

Banana

Mango }

Duplicate "Apple"

❌ Removed

Set Syntax

set_name = {item1, item2, item3}

Example

colors = {"Red", "Green", "Blue"}

Creating a Set

animals = {"Dog", "Cat", "Tiger"}

print(animals)

Output

{'Dog', 'Cat', 'Tiger'}

Duplicate Values are Removed

numbers = {1, 2, 2, 3, 4, 4, 5}

print(numbers)

Output

{1, 2, 3, 4, 5}

Finding the Length of a Set

fruits = {"Apple", "Banana", "Mango"}

print(len(fruits))

Output

3

Accessing Set Items

Since sets are **unordered**, you **cannot access items using indexes**.

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❌ Wrong

```
fruits = {"Apple", "Banana", "Mango"}


print(fruits[0])
```

Output

TypeError

✅ Correct Method

```
fruits = {"Apple", "Banana", "Mango"}


for fruit in fruits:
    print(fruit)
```

Adding an Item

```
fruits = {"Apple", "Banana"}


fruits.add("Mango")


print(fruits)
```

Output

{'Apple', 'Banana', 'Mango'}

Adding Multiple Items

```
fruits = {"Apple"}


fruits.update(["Banana", "Mango"])


print(fruits)
```

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Removing an Item

```
fruits = {"Apple", "Banana", "Mango"}


fruits.remove("Banana")


print(fruits)
```

Using discard()

```
fruits = {"Apple", "Banana"}


fruits.discard("Orange")


print(fruits)
```

Output

```
{'Apple', 'Banana'}
```

discard() does not give an error if the item is not present.

Using pop()

```
fruits = {"Apple", "Banana", "Mango"}


fruits.pop()


print(fruits)
```

pop() removes a random item because sets are unordered.

Clearing a Set

```
fruits = {"Apple", "Banana"}
```

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fruits.clear()

print(fruits)

Output

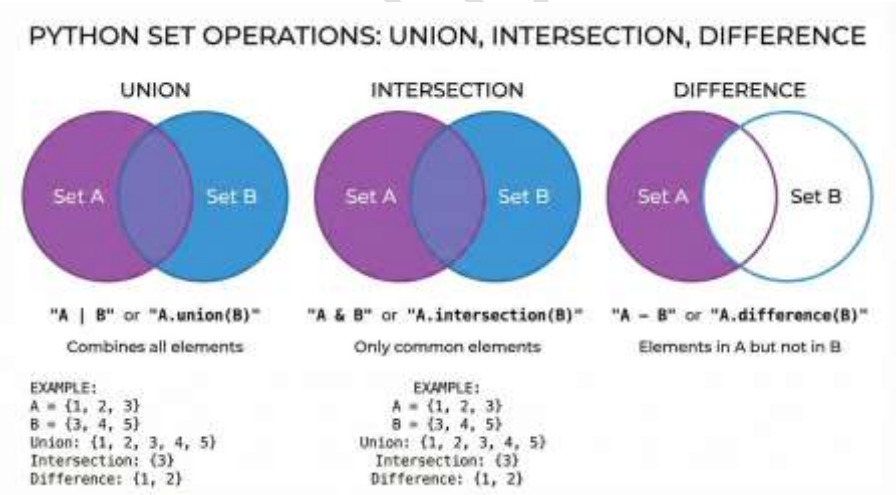
set()

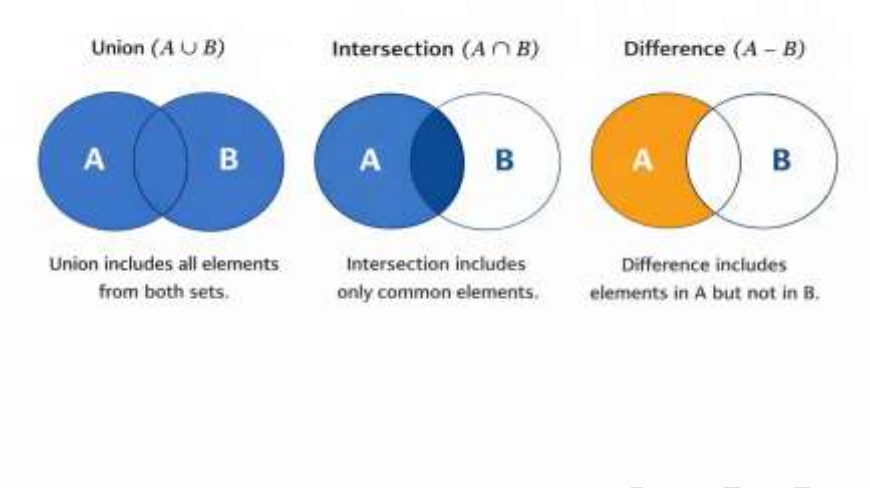
Deleting a Set

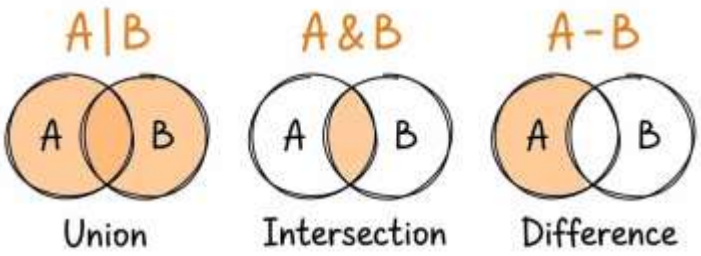
fruits = {"Apple", "Banana"}

del fruits

Set Operations







Union of Sets

Combines all unique items.

set1 = {1, 2, 3}

set2 = {3, 4, 5}

print(set1.union(set2))

Output

{1, 2, 3, 4, 5}

Intersection of Sets

Shows common items.

```
set1 = {1, 2, 3}

set2 = {2, 3, 4}


print(set1.intersection(set2))
```

Output

```
{2, 3}
```

Difference of Sets

Shows items present in the first set but not in the second.

```
set1 = {1, 2, 3}

set2 = {2, 4}


print(set1.difference(set2))
```

Output

```
{1, 3}
```

Symmetric Difference

Shows items that are in either set but not in both.

```
set1 = {1, 2, 3}

set2 = {3, 4, 5}


print(set1.symmetric_difference(set2))
```

Output

```
{1, 2, 4, 5}
```

Checking Membership

```
colors = {"Red", "Blue", "Green"}
```

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print("Blue" in colors)

Output

True

Loop Through a Set

colors = {"Red", "Blue", "Green"}

for color in colors:

  print(color)

Difference Between List, Tuple, Dictionary, and Set

Feature	List	Tuple	Dictionary	Set
Brackets	[]	()	{}	{}
Ordered	✅	✅	✅ (by insertion order)	❌
Duplicates	✅	✅	Keys ❌, Values ✅	❌
Mutable	✅	❌	✅	✅
Indexing	✅	✅	Keys	❌

Real-Life Examples

Unique Student Roll Numbers

roll_numbers = {101, 102, 103, 104}

Unique Subjects

subjects = {"Math", "Science", "English"}

Unique Sports

sports = {"Cricket", "Football", "Tennis"}

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Practical Session

Practical Exercise 1

```
colors = {"Red", "Blue", "Green"}


print(colors)
```

Practical Exercise 2

```
numbers = {10, 20, 20, 30, 30}


print(numbers)
```

Practical Exercise 3

```
animals = {"Dog", "Cat"}


animals.add("Tiger")


print(animals)
```

Practical Exercise 4

```
fruits = {"Apple", "Banana", "Mango"}


for fruit in fruits:
    print(fruit)
```

Practical Exercise 5

```
set1 = {1, 2, 3}
```

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```
set2 = {3, 4, 5}


print(set1.union(set2))
```

Mini Activity

Guess the Output

```
numbers = {1, 2, 2, 3}


print(numbers)
```

Answer

{1, 2, 3}

Guess the Output

```
colors = {"Red", "Blue"}


print("Blue" in colors)
```

Answer

True

Classroom Activity

Create a set of:

- 5 Fruits
- 5 Animals
- 5 Colors

Then:

1. Add one more item.
2. Remove one item.

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3. Print all items using a for loop.
4. Create another set and find the **union** and **intersection**.

Quiz

Choose the Correct Answer

1. Which brackets are used for sets?

A) []

B) ()

C) {}

D) <>

✅ **Answer:** C

2. Can a set store duplicate values?

A) Yes

B) No

✅ **Answer:** B

3. Which method adds a single item to a set?

A) append()

B) insert()

C) add()

D) push()

✅ **Answer:** C

4. Which method combines two sets?

A) join()

B) union()

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C) merge()

D) append()

✅ **Answer:** B

5. Can we access a set using an index like set[0]?

A) Yes

B) No

✅ **Answer:** B

Homework

1. Create a set of your favorite fruits.
2. Add one new fruit using add().
3. Remove one fruit using remove().
4. Create another fruit set and find the union().
5. Find the intersection() of two sets.
6. Check if "Apple" is present in the set.
7. Print all items using a for loop.
8. Create a set with duplicate numbers and observe the output.
9. Find the length of a set using len().
10. Explain why sets are useful in real life.

Key Points to Remember

- A **Set** is a collection of **unique items**.
- Sets use **curly braces {}**.
- Sets **do not allow duplicate values**.
- Sets are **unordered**, so they do not support indexing.
- Use add() to add an item and remove() or discard() to remove items.

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- Common operations include union(), intersection(), difference(), and symmetric_difference().
- Sets are very useful when you need to remove duplicates or work with unique collections of data.

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